YEAR 9 PHYSICS, AQA GCSE, UNIT 2, PART 1

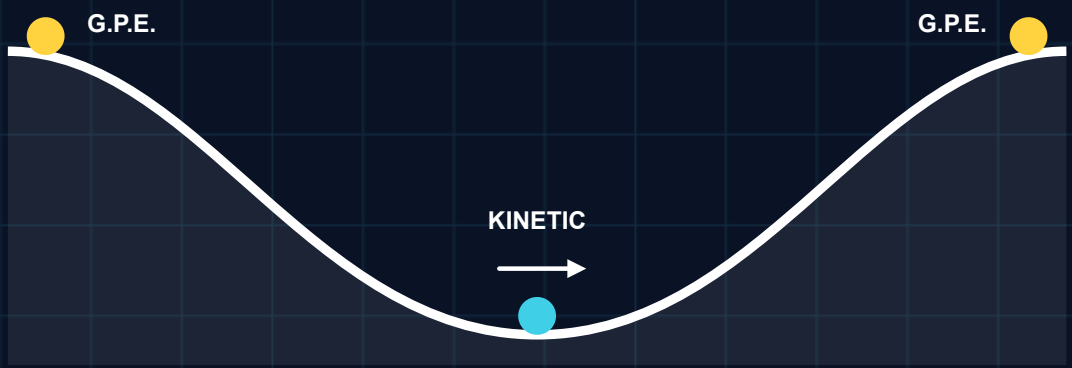
ENERGY STORES AND TRANSFERS

PART 1 OF 3

LESSONS 1–3: ENERGY STORES AND SYSTEMS, THE KINETIC STORE, THE GRAVITATIONAL POTENTIAL STORE

3 LESSONS **1** PRACTICAL **3** NEW EQUATIONS

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

HOW THIS BOOKLET WORKS

Energy is never made and never destroyed: it only moves from one store to another. This booklet shows you how to follow it, and how to calculate it. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try “Your turn” alone.

TRY IT
Practice that builds up: first with support, then on your own. Each lesson ends with the slides' Quick check.

PRACTICAL
Record your measurements and work out results from them.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: Energy is measured in ___joules___ **(joules / newtons / watts)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of energy is **the watt** | **(the joule)** | **the newton** .

THE ROUTE THROUGH THIS PART

#	LESSON	WEEK
01	**Energy stores and systems** The energy stores, four ways to transfer energy, systems	1
02	**The kinetic store** $E_k = \frac{1}{2} m v^2$; grams to kilograms; double the speed, four times the energy	1
03	**The gravitational potential store** What g means, $E_p = m g h$, height gained, finding h or m	2

This booklet is Part 1 of 3: Lessons 1–3. The **equation toolkit** on the last page has every equation you have met so far, with formula triangles to help you rearrange.

Energy: Stores and Transfers, Year 9 workbook Part 1 (ANSWERS), © 2026 PanPhy Labs. All rights reserved.

Lesson

01 ENERGY STORES AND SYSTEMS

AQA Physics 4.1.1.1; Combined Science 6.1.1.1

BIG QUESTION **A skateboarder rolls down one side of a ramp and up the other, but never quite gets back to the starting height. Where did the energy go?**

BY THE END YOU CAN

- name the energy stores and give an example of each
- name the four ways energy is transferred
- explain what a system is
- describe the energy transfers when something changes

energy store **transfer** **system** **closed system** **joule (J)**

YOU WILL NEED

This booklet, a pencil and a ruler.

REMEMBER

WHAT DO YOU ALREADY THINK?

No wrong answers: this is your starting point.

1

Name something that has energy. How can you tell?

For example, a moving car (it is moving) or a hot drink (it is hot). Any sensible answer.

2

Energy is measured in **joules** **(joules / newtons / metres)**.

joules (J)

3

Rub your hands together. They get **warmer** **(warmer / colder)**. Why?

Warmer: rubbing (friction) transfers energy to the thermal store of your hands.

EXPLAIN

STORES AND TRANSFERS

Energy is never made and never used up. It is **stored**, and it is **transferred** (moved) from one store to another. Energy is measured in **joules (J)**.

THE FIVE MAIN ENERGY STORES

STORE	ENERGY IS STORED BECAUSE...	EXAMPLE
Kinetic	the object is moving	a rolling football
Gravitational potential	the object is raised up	a book on a high shelf
Elastic potential	the object is stretched or squashed	a stretched spring, a drawn bow
Thermal	the object is hot	a cup of tea
Chemical	of the substances it is made of	food, fuel, a battery

You may also meet the **magnetic** store (two magnets held apart), the **electrostatic** store (a charged balloon) and the **nuclear** store (uranium fuel), but GCSE questions rarely use them.

FOUR WAYS TO TRANSFER ENERGY

TRANSFERRED...	WHAT HAPPENS	EXAMPLE
mechanically	a force moves something	pushing a trolley
electrically	a current flows	a battery lighting a lamp
by heating	a hotter object warms a colder one	a hob heating a pan
by waves	a wave carries it: light, infrared (both called radiation) or sound	the Sun warming your face; a loudspeaker

! **Stores hold energy. Transfers move it.** Light, sound and electricity are **not** stores: they are ways energy travels from one store to another.

SYSTEMS

A **system** is the object, or group of objects, that we choose to study. In a **closed system**, no energy gets in or out, so the **total** energy stays the same. The energy only moves between stores.

DESCRIBE ANY CHANGE IN THREE STEPS

1. Which store **empties**?
2. Which store **fills**?
3. **How** is the energy transferred: mechanically, electrically, by heating or by waves?

WITH YOUR TEACHER SLIDE 7

A car brakes and stops. Describe the energy transfer.

EMPTIES	the kinetic store of the car
FILLS	the thermal store of the brakes
HOW	mechanically: friction is a force
WRITE	Energy is transferred mechanically from the kinetic store of the car to the thermal store of the brakes.

YOUR TURN

A wind-up toy car is let go and moves off. Describe the energy transfer.

EMPTIES	the elastic potential store of the spring
FILLS	the kinetic store of the car
HOW	mechanically: the spring's force turns the wheels
WRITE	Energy is transferred mechanically from the elastic potential store of the spring to the kinetic store of the car.

PRACTISE

TRY IT

1. Draw a line from each store to its example.

Kinetic	a stretched bow
Gravitational potential	a hot bath
Elastic potential	a rolling ball
Chemical	a battery
Thermal	a book on a high shelf

2. Draw a ring around the correct answer in each box.

Light is **an energy store** / **a way energy is transferred**

A battery stores energy in its **chemical store** / **kinetic store** / **thermal store**

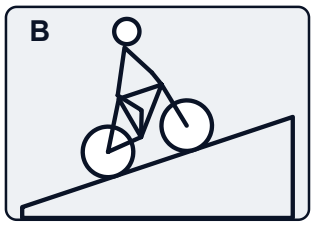
At the top of its flight, a ball has most energy in its

kinetic store / **gravitational potential store**

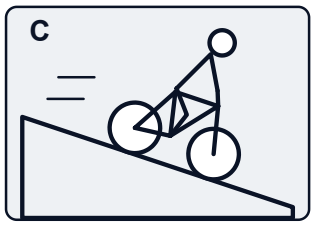
3 The pictures tell a story. For each picture, write the store that **empties** and the store that **fills**. Choose from the words under each gap.



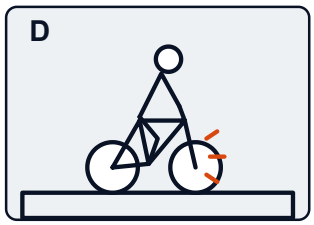
Eating breakfast



Cycling up the hill



Freewheeling down



Braking at the bottom

PICTURE	STORE THAT EMPTIES	STORE THAT FILLS
A Eating	chemical (in the food)	chemical (in the body)
B Cycling up	chemical / kinetic **chemical (body)**	gravitational potential / elastic potential **gravitational potential (and thermal)**
C Freewheeling down	gravitational potential / chemical **gravitational potential**	kinetic / nuclear **kinetic**
D Braking	kinetic / thermal **kinetic**	thermal / chemical **thermal (brakes and surroundings)**

4 SUPPORTED Complete each sentence. The choices are after each gap.

A ball is thrown upwards. Energy is transferred from the **kinetic** **(kinetic / chemical)** store to the **gravitational potential** **(gravitational potential / thermal)** store.

A fireplace warms a person. Energy is transferred by infrared waves from the **chemical** **(chemical / kinetic)** store of the fuel to the **thermal** **(thermal / kinetic)** store of the person.

A child lands on a trampoline. Energy is transferred mechanically from the child's **kinetic** **(kinetic / chemical)** store to the trampoline's **elastic potential** **(elastic potential / thermal)** store.

5 ON YOUR OWN A bungee jumper falls, and the cord stretches until the jumper stops at the bottom. Describe the energy transfers. Use the three steps.

As the jumper falls, energy is transferred from the gravitational potential store of the jumper to the kinetic store. As the cord stretches, energy is transferred from the kinetic store to the elastic potential store of the cord.

6 Explain what is wrong with what Sam says.

Sam When the car stopped, its energy was used up.

Energy cannot be used up (destroyed). The energy in the car's kinetic store was transferred to the thermal store of the brakes and the surroundings.

CHECKPOINT

QUICK CHECK

Slide 10. Answer each question, then check it as your teacher reveals the answer.

1
What is the unit of energy?

The joule (J).

2
Is light an energy store?

No: it is a way energy is transferred (by waves).

3
A ball is thrown up. Which store fills?

Its gravitational potential store.

4
A car brakes. Which store fills?

The thermal store of the brakes.

5
What is a closed system?

No energy gets in or out, so the total stays the same.

6
“The energy was used up.” Correct this sentence.

The energy was transferred to the thermal store of the surroundings.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** Name both stores and say which one empties and which one fills.

7 A student throws a ball straight up. Complete the sentence. **[2 marks]**

As the ball rises, energy is transferred from the **kinetic** **(kinetic / chemical / thermal)** store of the ball to its **gravitational potential** **(elastic potential / gravitational potential / nuclear)** store.

8 What is the unit of energy? Tick **one** box. **[1 mark]**

- [x] joule
- [ ] newton
- [ ] watt
- [ ] kilogram

9 A child lands on a trampoline. Describe the energy transfer as the mat and springs stretch and the child slows down. **[2 marks]**

Energy is transferred mechanically from the kinetic store of the child (1) to the elastic potential store of the trampoline (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Energy is measured in **joules (J)**. It is never made or destroyed.
- Main stores: kinetic, gravitational potential, elastic potential, thermal, chemical.
- Transfers: mechanically, electrically, by heating, by waves.
- In a closed system, the total energy stays the same.

COVER THE PAGE AND ANSWER

1 What is the unit of energy?

2 Name four energy stores.

3 Is light a store or a way of transferring energy?

4 What is a closed system?

I CAN…

- name the energy stores
- name the four ways energy is transferred
- explain what a system is
- describe energy transfers using “from … to …”

Lesson

02 THE KINETIC STORE

AQA Physics 4.1.1.2; Combined Science 6.1.1.2

BIG QUESTION **A car travels twice as fast. Does it have twice as much energy in its kinetic store, or more?**

BY THE END YOU CAN

- say what the kinetic store depends on
- calculate kinetic energy with $E_k = \frac{1}{2} m v^2$
- convert grams to kilograms
- explain why doubling the speed gives four times the energy

kinetic store | mass (kg) | speed (m/s) | squared

YOU WILL NEED

Calculator. Practical: trolley, 1.00 m track, stopwatch, balance.

REMEMBER

DO NOW

From earlier lessons. No notes!

1

Which store does a moving object have energy in?

The kinetic store.

2

A car brakes. Which store fills?

The thermal store (of the brakes and the surroundings).

3

Light is _____ (a store / a way energy is transferred).

a way energy is transferred

A way energy is transferred (by waves).

EXPLAIN

ENERGY IN A MOVING OBJECT

Anything that moves has energy in its **kinetic store**. The more **mass** it has, and the faster it moves, the more energy it has.

$$E_k = \frac{1}{2} m v^2$$

kinetic energy = 0.5 × mass × (speed)²

E_k in joules (J), m in kilograms (kg), v in metres per second (m/s)

Only the speed is squared. In this unit you use the equation to calculate the energy.

! **Square the speed first:** $v^2 = v \times v$. Then multiply by the mass and by 0.5.

WITH YOUR TEACHER SLIDE 12

A 1200 kg car moves at 15 m/s. Calculate the energy in its kinetic store.

EQUATION	$E_k = \frac{1}{2} m v^2$
SUBSTITUTE	$E_k = 0.5 \times 1200 \times 15^2$
ANSWER	$E_k = 135\,000$ J

YOUR TURN

A 900 kg car moves at 20 m/s. Calculate the energy in its kinetic store.

EQUATION	$E_k = \frac{1}{2} m v^2$
SUBSTITUTE	$E_k = 0.5 \times 900 \times 20^2$
ANSWER	$E_k = 180\,000$ J

MASS IN GRAMS? CONVERT FIRST

The mass must be in kilograms. Divide grams by 1000. For example, $600\text{ g} = 0.600\text{ kg}$.

WITH YOUR TEACHER SLIDE 12

A 450 g football is kicked at 20 m/s. Calculate the energy in its kinetic store.

CONVERT	$m = 450\text{ g} = 0.450\text{ kg}$
EQUATION	$E_k = \frac{1}{2} m v^2$
SUBSTITUTE	$E_k = 0.5 \times 0.450 \times 20^2$
ANSWER	$E_k = 90$ J

YOUR TURN

A 300 g ball is thrown at 12 m/s. Calculate the energy in its kinetic store.

CONVERT	$m = 300\text{ g} = 0.300\text{ kg}$
EQUATION	$E_k = \frac{1}{2} m v^2$
SUBSTITUTE	$E_k = 0.5 \times 0.300 \times 12^2$
ANSWER	$E_k = 21.6$ J

DOUBLE THE SPEED, FOUR TIMES THE ENERGY

SPEED OF A 1200 KG CAR	v^2	E_k
15 m/s	225	135 000 J
30 m/s (2 × faster)	900	540 000 J (4 × more)
45 m/s (3 × faster)	2025	1 215 000 J (9 × more)

Twice the speed gives $2^2 = 4$ times the energy. Three times the speed gives $3^2 = 9$ times. This is why a crash at high speed does so much more damage, and why the brakes must transfer so much more energy to stop the car. Doubling the **mass** only doubles the energy, because the mass is not squared.

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

Doubling the mass makes the kinetic energy

stay the same	**2 times bigger**	4 times bigger

Doubling the speed makes the kinetic energy

stay the same	2 times bigger	**4 times bigger**

2 SUPPORTED A 2.0 kg ball rolls at 3.0 m/s. Calculate the energy in its kinetic store.

EQUATION	$E_k = \frac{1}{2} m v^2$
SUBSTITUTE	$E_k = 0.5 \times 2.0 \times 3.0^2$
ANSWER	**9.0 J** unit:

3 **SUPPORTED** A 400 g ball is kicked at 15 m/s. Calculate the energy in its kinetic store. *Convert the mass first!*

CONVERT	**m = 400 ÷ 1000 = 0.400 kg**
EQUATION	$E_k = \frac{1}{2} m v^2$
SUBSTITUTE	$E_k = 0.5 \times 0.400 \times 15^2$
ANSWER	**45 J** unit:

4 Complete the table. Show your working underneath.

Object	Mass	Speed	E_k / J
Lorry	**30 000 kg**	**2.0 m/s**	**60 000**
Cyclist and bike	**80 kg**	**6.0 m/s**	**1440**
Tennis ball	**58 g**	**50 m/s**	**72.5**
Bullet	**20 g**	**300 m/s**	**900**

5 **ON YOUR OWN** A 0.16 kg cricket ball is bowled at 40 m/s. Calculate the energy in its kinetic store.

$E_k = 0.5 \times 0.16 \times 40^2 = 128$ J

6 **CHALLENGE** A car speeds up from 10 m/s to 30 m/s. How many times bigger is the energy in its kinetic store? Explain without calculating the energy.

9 times bigger. The speed is 3 times bigger, and the speed is squared: $3^2 = 9$.

7 Explain what is wrong with what Ava says.

Ava: At 4 m/s my bike has twice as much kinetic energy as at 2 m/s.

The speed is squared, so doubling the speed gives $2^2 = 4$ times as much kinetic energy, not twice as much.

INVESTIGATE

MEASURE THE ENERGY OF A MOVING TROLLEY

Work in a group of three: one pushes, one times, one records.

1. Measure the trolley's mass on the balance in grams. Convert it to kilograms.
2. Mark a 1.00 m section of the track with tape.
3. Give the trolley a gentle push so it rolls through the section. Time how long it takes to cross the 1.00 m.
4. Repeat twice more with the same push, and find the mean time.
5. Calculate the speed, then the kinetic energy.

8. Record your results.

Mass / g	Mass / kg	Time 1 / s	Time 2 / s	Time 3 / s	Mean time / s
500	0.500	1.62	1.55	1.59	1.59

9. Calculate the speed: $\text{speed} = \dfrac{\text{distance}}{\text{time}}$.

EQUATION	$v = \dfrac{s}{t}$
SUBSTITUTE	$v = \dfrac{1.00}{1.59}$
ANSWER	**0.63 m/s (sample)** unit:

10. Calculate the energy in the trolley's kinetic store.

EQUATION	$E_k = \frac{1}{2}\, m\, v^2$
SUBSTITUTE	$E_k = 0.5 \times 0.500 \times 0.63^2$
ANSWER	**0.10 J (sample)** unit:

11. Which measurement has the biggest effect on your answer? **the mass** / **the time**

Explain why.

The time: the speed is worked out from it, and the speed is squared, so any error in the time is made bigger.

CHECKPOINT

QUICK CHECK

Slide 15. Answer each question, then check it as your teacher reveals the answer.

1
A 60 kg runner moves at 5.0 m/s. Calculate the energy in their kinetic store.

$0.5 \times 60 \times 5.0^2 = 750$ J

2
Convert 250 g to kilograms.

0.250 kg

3
The speed doubles. What happens to the kinetic energy?

It is multiplied by 4 ($\times 4$).

4
The speed halves. What happens to the kinetic energy?

It becomes a quarter.

5
A 10 g bullet moves at 400 m/s. Calculate its kinetic energy.

$0.5 \times 0.010 \times 400^2 = 800$ J

6
The mass doubles. What happens to the kinetic energy?

It is multiplied by 2 ($\times 2$).

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Write the equation, then substitute, then give the answer with its unit.

12 A cyclist and bike have a total mass of 70 kg and move at 8.0 m/s. Calculate the energy in their kinetic store. **[2 marks]**

EQUATION	$E_k = \frac{1}{2} m v^2$
SUBSTITUTE	$E_k = 0.5 \times 70 \times 8.0^2$ **(1)**
ANSWER	**2240 J (1)** unit:

13 Car A and car B have the same mass. Car B moves twice as fast as car A. Complete the sentence. **[2 marks]**

Car B has **4** **(2 / 4 / 8)** times as much energy in its kinetic store as car A, because in the equation the speed is **halved** / **squared** / **doubled**.

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- $E_k = \frac{1}{2}\,m\,v^2$: mass in kg, speed in m/s, energy in J.
- Convert grams to kilograms: divide by 1000.
- Only the speed is squared: 2 × the speed gives 4 × the energy.
- 2 × the mass gives 2 × the energy.

COVER THE PAGE AND ANSWER

1 What two things does the kinetic store depend on?

2 Write the equation for kinetic energy.

3 Convert 350 g to kilograms.

4 The speed triples. What happens to the kinetic energy?

I CAN…

- calculate kinetic energy with $E_k = \frac{1}{2}\,m\,v^2$
- convert grams to kilograms
- explain why doubling the speed gives four times the energy

Lesson

03 THE GRAVITATIONAL POTENTIAL STORE

AQA Physics 4.1.1.2; Combined Science 6.1.1.2

BIG QUESTION **Carrying a bag up the stairs is hard work. Where does that energy go, and can you get it back?**

BY THE END YOU CAN

- explain what $g = 9.8$ N/kg means
- calculate energy with $E_p = m\,g\,h$
- use the height gained
- rearrange to find a height or a mass

gravitational potential store

gravitational field strength **newton (N)**

height gained

ELECTRIC MOUNTAIN

At Dinorwig in Wales, water is pumped up into a lake high on a mountain. When the country suddenly needs electricity, the water rushes down through turbines. The energy waits in the water's gravitational potential store until it is needed.

REMEMBER

DO NOW

From earlier lessons. No notes!

1

In $E_k = \frac{1}{2}\,m\,v^2$, what does v stand for? Give its unit.

Speed, in metres per second (m/s).

2

Convert 600 g to kilograms.

0.600 kg

3

A 2.0 kg ball moves at 4.0 m/s. Calculate its kinetic energy.

$E_k = 0.5 \times 2.0 \times 4.0^2$

$E_k = 16$ J

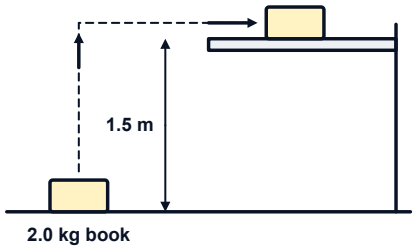
EXPLAIN

ENERGY FROM HEIGHT

Lift an object up and you transfer energy to its **gravitational potential store**. Let it fall and the energy is transferred to its kinetic store. The heavier the object and the higher you lift it, the more energy it stores.

WHAT g MEANS

! On Earth, gravity pulls every kilogram down with a force of **9.8 newtons**. We write this as $g = 9.8$ N/kg (newtons per kilogram). It is called the **gravitational field strength**. On the Moon, $g = 1.6$ N/kg, so the same lift stores less energy there.

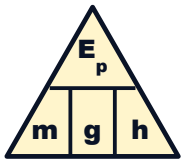


$$E_p = m\,g\,h$$

gravitational potential energy = mass × gravitational field strength × height

E_p in J, m in kg, g in N/kg, h in m

Rearranged: $h = \dfrac{E_p}{m\,g}$ and $m = \dfrac{E_p}{g\,h}$



WITH YOUR TEACHER — SLIDE 20

A 2.0 kg book is lifted onto a shelf 1.5 m high. Calculate the energy transferred to its gravitational potential store.

EQUATION	$E_p = m\,g\,h$
SUBSTITUTE	$E_p = 2.0 \times 9.8 \times 1.5$
ANSWER	$E_p = 29.4$ J

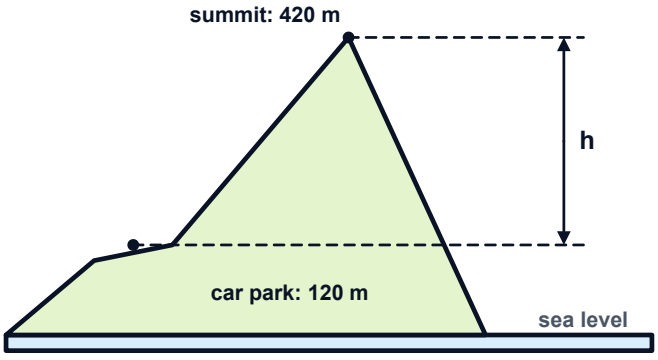
YOUR TURN

A 10 kg box is lifted onto a table 0.80 m high. Calculate the energy transferred to its gravitational potential store.

EQUATION	$E_p = m\,g\,h$
SUBSTITUTE	$E_p = 10 \times 9.8 \times 0.80$
ANSWER	$E_p = 78.4$ J

USE THE HEIGHT GAINED

Only the **change** in height matters. A hiker climbs from a car park 120 m above sea level to a summit 420 m above sea level. Use the height gained, the difference between the two heights, not the height of the summit.



WITH YOUR TEACHER SLIDE 21

A 55 kg hiker climbs from 120 m to 420 m above sea level. Calculate the energy transferred to the hiker's gravitational potential store.

HEIGHT	$h = 420 - 120 = 300$ m
EQUATION	$E_p = m\,g\,h$
SUBSTITUTE	$E_p = 55 \times 9.8 \times 300$
ANSWER	$E_p = 161\,700$ J

YOUR TURN

A 40 kg child climbs from 50 m to 250 m above sea level. Calculate the energy transferred to the child's gravitational potential store.

HEIGHT	$h = 250 - 50 = 200$ m
EQUATION	$E_p = m\,g\,h$
SUBSTITUTE	$E_p = 40 \times 9.8 \times 200$
ANSWER	$E_p = 78\,400$ J

FINDING THE HEIGHT OR THE MASS

WITH YOUR TEACHER SLIDE 22

A 60 kg climber gains 17 640 J in their gravitational potential store. How high did they climb?

EQUATION	$E_p = m\,g\,h$
REARRANGE	$h = \dfrac{E_p}{m\,g}$
SUBSTITUTE	$h = \dfrac{17\,640}{60 \times 9.8}$
ANSWER	h = 30 m

YOUR TURN

An 80 kg climber gains 15 680 J in their gravitational potential store. How high did they climb?

EQUATION	$E_p = m\,g\,h$
REARRANGE	$h = \dfrac{E_p}{m\,g}$
SUBSTITUTE	$h = \dfrac{15\,680}{80 \times 9.8}$
ANSWER	h = 20 m

WITH YOUR TEACHER SLIDE 22

A crate is lifted 2.0 m and gains 392 J. Calculate its mass.

EQUATION	$E_p = m\,g\,h$
REARRANGE	$m = \dfrac{E_p}{g\,h}$
SUBSTITUTE	$m = \dfrac{392}{9.8 \times 2.0}$
ANSWER	m = 20 kg

YOUR TURN

A sack is lifted 3.0 m and gains 735 J. Calculate its mass.

EQUATION	$E_p = m\,g\,h$
REARRANGE	$m = \dfrac{E_p}{g\,h}$
SUBSTITUTE	$m = \dfrac{735}{9.8 \times 3.0}$
ANSWER	m = 25 kg

Get the marks: Work out the bottom of the fraction first ($60 \times 9.8 = 588$), then divide. Or put brackets round it on your calculator.

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

On the Moon, lifting a box 1 m stores

more energy than on Earth	**less energy than on Earth**
the same energy as on Earth	

Lifting a box twice as high makes the energy stored

half as big	**twice as big**	**four times as big**

2 SUPPORTED A 0.50 kg ball is lifted 3.0 m. Calculate the energy transferred to its gravitational potential store.

EQUATION	$E_p = m\,g\,h$	
SUBSTITUTE	$E_p = 0.50 \times 9.8 \times 3.0$	
ANSWER	**14.7 J**	unit:

3 SUPPORTED A 600 g book is lifted 1.2 m onto a shelf. Calculate the energy transferred. *Convert the mass first!*

CONVERT	**m = 600 ÷ 1000 = 0.600 kg**	
EQUATION	$E_p = m\,g\,h$	
SUBSTITUTE	$E_p = 0.600 \times 9.8 \times 1.2$	
ANSWER	**7.1 J**	unit:

4 Complete the table. Use $g = 9.8$ N/kg. Show your working underneath.

Object	Mass / kg	Height gained / m	E_p / J
Child at the top of a slide	30	2.5	735
Rock on a cliff	4.0	5.0	196
Crate on a lorry	30	3.0	882
Lift with passengers	800	12	94 080

5 **CHALLENGE** A 50 kg student gains 4.9 kJ in their gravitational potential store climbing stairs. How high did they climb? *Convert kJ to J first.*

$E_p = 4900$ J; $h = \dfrac{E_p}{m\,g} = \dfrac{4900}{50 \times 9.8} = 10$ m

INVESTIGATE

ACTIVITY: LIFT IT

Work in pairs. Lift gently and keep the floor clear.

Choose three objects. Measure the mass of each and the height of the shelf or bench you lift it onto. Calculate the energy transferred to each gravitational potential store.

6 Record and calculate.

Object	Mass / kg	Height / m	E_p / J
textbook	1.2	0.90	10.6
water bottle	0.50	0.90	4.4
school bag	4.0	0.90	35.3

7 How high would you need to lift your lightest object so that it stores the same energy as your heaviest object?

Sample: $h = \dfrac{E_p}{m\,g} = \dfrac{35.3}{0.50 \times 9.8} = 7.2$ m

CHECKPOINT

QUICK CHECK

Slide 23. Answer each question, then check it as your teacher reveals the answer.

1

What does $g = 9.8$ N/kg mean?

Each kilogram is pulled down with 9.8 N.

2

A 5.0 kg mass is lifted 2.0 m. Calculate the energy transferred.

$5.0 \times 9.8 \times 2.0 = 98$ J

3

A 0.50 kg ball gains 24.5 J. Calculate the height it was lifted.

$$\frac{24.5}{0.50 \times 9.8} = 5.0 \text{ m}$$

4

The same lift is done on the Moon. More, less or the same energy?

Less energy: g is smaller.

5

Something goes from 30 m to 80 m up. What is h?

50 m

6

The height doubles. What happens to the energy?

It is multiplied by 2 ($\times 2$).

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** g is always given in the question. AQA uses 9.8 N/kg.

8 A crane lifts a 250 kg load through a height of 12 m. Calculate the energy transferred to the load's gravitational potential store. Gravitational field strength = 9.8 N/kg **[2 marks]**

EQUATION	$E_p = m\,g\,h$
SUBSTITUTE	$E_p = 250 \times 9.8 \times 12$ **(1)**
ANSWER	**29 400 J (1)** unit:

9 A 0.40 kg ball gains 7.84 J in its gravitational potential store when it is thrown upwards. Calculate how high it rises. Gravitational field strength = 9.8 N/kg **[3 marks]**

EQUATION	$E_p = m\,g\,h$
REARRANGE	$h = \dfrac{E_p}{m\,g}$ **(1)**
SUBSTITUTE	$h = \dfrac{7.84}{0.40 \times 9.8}$ **(1)**
ANSWER	**2.0 m (1)** unit:

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- $g = 9.8$ N/kg: gravity pulls each kilogram down with 9.8 N.
- $E_p = m\,g\,h$, with the height gained in metres.
- Rearranged: $h = \dfrac{E_p}{m\,g}$ and $m = \dfrac{E_p}{g\,h}$.
- Twice the height or twice the mass gives twice the energy.

COVER THE PAGE AND ANSWER

1 What does $g = 9.8$ N/kg mean?

2 Write the equation for gravitational potential energy.

3 A 3.0 kg bag is lifted 1.0 m. How much energy does it gain?

4 Rearrange $E_p = m\,g\,h$ to find h.

I CAN…

- explain what $g = 9.8$ N/kg means
- calculate $E_p = m\,g\,h$ using the height gained
- rearrange $E_p = m\,g\,h$ to find a height or a mass

REFERENCE

KEY WORDS

Learn the meaning and the spelling. In exams, precise words earn marks: say “transferred to the thermal store of the surroundings”, never “lost”.

Energy store L1: Where energy is held. The main stores are kinetic, gravitational potential, elastic potential, thermal and chemical

Joule (J) L1: The unit of energy

System L1: The object or group of objects that we choose to study

Closed system L1: A system that no energy enters or leaves; its total energy stays the same

Kinetic store L2: Energy an object has because it is moving

Gravitational potential store L3: Energy an object has because it is raised in a gravitational field

Gravitational field strength (g) L3: The force of gravity on each kilogram: 9.8 N/kg on Earth

FINISH

PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

I CAN…	R / A / G
01: Energy stores and systems	
name the energy stores	
name the four ways energy is transferred	
explain what a system is	
describe energy transfers using “from … to …”	
02: The kinetic store	
calculate kinetic energy with $E_k = \frac{1}{2} m v^2$	
convert grams to kilograms	
explain why doubling the speed gives four times the energy	
03: The gravitational potential store	
explain what $g = 9.8$ N/kg means	
calculate $E_p = m g h$ using the height gained	
rearrange $E_p = m g h$ to find a height or a mass	

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

LESSON 1: ENERGY STORES AND SYSTEMS

DO NOW

1. For example, a moving car (it is moving) or a hot drink (it is hot). Any sensible answer.
2. joules (J)
3. Warmer: rubbing (friction) transfers energy to the thermal store of your hands.

COVER THE PAGE

1. Joule (J).
2. Any four of: kinetic, gravitational potential, elastic potential, thermal, chemical.
3. A way of transferring energy (by waves).
4. A system that no energy enters or leaves, so its total energy stays the same.

LESSON 2: THE KINETIC STORE

DO NOW

1. The kinetic store.
2. The thermal store (of the brakes and the surroundings).
3. A way energy is transferred (by waves).

COVER THE PAGE

1. Mass and speed.
2. $E_k = \frac{1}{2}\,m\,v^2$
3. 0.350 kg
4. It becomes 9 times bigger.

LESSON 3: THE GRAVITATIONAL POTENTIAL STORE

DO NOW

1. Speed, in metres per second (m/s).
2. 0.600 kg
3. $E_k = 0.5 \times 2.0 \times 4.0^2$
   $E_k = 16$ J

COVER THE PAGE

1. Gravity pulls each kilogram down with a force of 9.8 N.
2. $E_p = m\,g\,h$
3. 29.4 J
4. $h = \dfrac{E_p}{m\,g}$

REFERENCE

EQUATION TOOLKIT

Every equation from Lessons 1–3. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it. Use $g = 9.8\ \text{N/kg}$.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find h from $E_p = m\,g\,h$, cover h. E_p is above m and g, so $h = \dfrac{E_p}{m\,g}$. Kinetic energy has no triangle: in this unit you only use it to calculate the energy.

$E_k = \frac{1}{2}\,m\,v^2$

Lesson 2

kinetic energy = 0.5 × mass × (speed)²

J, kg, m/s

Use it to calculate the energy. Only the speed is squared.

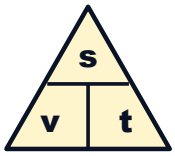
$s = v\,t$

Lesson 2

distance travelled = speed × time

m, m/s, s

$v = \dfrac{s}{t}$ and $t = \dfrac{s}{v}$



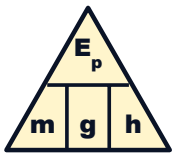
$E_p = m\,g\,h$

Lesson 3

gravitational potential energy = mass × gravitational field strength × height

J, kg, N/kg, m

$h = \dfrac{E_p}{m\,g}$ and $m = \dfrac{E_p}{g\,h}$



1

Write what you know. Convert first: g → kg, cm → m, kJ → J.

2

Write the equation in symbols.

3

Rearrange if you need to, then substitute the numbers.

4

Give the answer with a unit. Ask: is it sensible?